



If I look at the system level, the challenges are quite clear:

- Reducing cost, weight, and overall system size
- Simplifying connectors and wiring harnesses
- Improving efficiency and reliability
- Ensuring designs are truly future-ready

In a typical 12V system, as more features are added, battery load keeps increasing, driving higher circuit complexity. Higher current demand increases cost, weight, and size, which are primary limitations of the existing architecture. Connector design further compounds this. Currents in the range of 3 to 5 amps require larger pins and copper-heavy wiring harnesses, adding both weight and cost. Efficiency suffers, heating increases, and reliability becomes a concern. Wherever there is scope to improve reliability, it should be actively considered.

This brings me to a broader question: Are our designs really future-ready? As battery systems evolve, the rest of the vehicle architecture should evolve with them. In modern electric vehicles, especially two-wheelers and three-wheelers, the traction or motor controller is already predominantly powered by 48V. This is true for most scooters and three-wheelers on Indian roads today. Even in cars, systems like H-boxes or ISG are moving to 48V. Yet, despite this shift, the 12V battery continues to remain in the system.

Lighting losses: The 12V bottleneck

Instead of reviewing all 12V sub-systems, let me take lighting as an example. It is no longer just basic on and off. Modern lighting is becoming increasingly sophisticated, with blinking, animation, and dynamic behaviour that bring vehicles to life. In a typical two-wheeler, the front lighting system alone draws nearly 40W across high beam, low beam, DRL (daytime running light), and

indicators. These loads are usually driven by LED drivers using a non-synchronous buck-boost topology involving a diode and a MOSFET, and this is where efficiency begins to drop, often approaching 75%.

In conventional architectures, power comes either directly from a 12V battery or from a 48V-to-12V converter. The initial 48V-to-12V conversion itself introduces roughly a 10% loss. A DC-DC converter then reduces 48V to about 14V for the LED driver, while the buck-boost topology compensates for variations in LED forward voltage. When these stages are combined, overall efficiency drops to nearly 77%. In practical terms, almost one-quarter of total power is lost purely in conversion.

Looking more closely, about 25% of the loss comes from circuit conversion, with wiring harness and protection elements such as input diodes bringing the total losses close to 30%. Nearly one-third of battery energy is not used effectively.

A native 48V architecture offers a more efficient path. Instead of stepping down to 14V, a constant-voltage synchronous buck converter can generate an intermediate voltage of 36 to 40V. This matches typical LED string requirements, where high- and low-beam voltages are around 15V each and remain below 30V even in series, enabling more efficient direct driving. Replacing diodes with MOSFETs reduces conduction losses, and techniques such as zero-voltage switching minimise switching losses, allowing the first stage to reach around 92% efficiency.

This intermediate stage can then feed multiple constant-current synchronous buck stages for LED driving, each also operating above 92% efficiency. Combined, overall system efficiency rises to roughly 84.5%, a gain of about 7.5% over conventional approaches. Further reductions in harness, connector, and protection losses can add another 5%. Altogether, moving from a 12V architecture to a 48V system can deliver a

net efficiency gain of 12-13%, making a strong case for rethinking power delivery in modern vehicles.

From lighting to intelligence: The LCM concept

This naturally leads to the concept of an LCM, or lighting control module, a new generation of lighting systems that go far beyond conventional 'dumb' controllers by integrating intelligence, communication, and centralised control. An LCM typically operates with a single 48V power input and a CAN or LIN (controller area network or local interconnect network) interface that governs all lighting functions. High beam, low beam, daytime running lights, and indicators can all be controlled through a single communication port, significantly simplifying the system.

This approach reduces input harness complexity. Instead of multiple connectors and wiring paths, the system can use a single connector with just four pins, two for power and two for communication, while multiple outputs from the same PCB drive individual lighting loads. A single constant-voltage buck stage generates an intermediate 36V to 40V rail, which feeds multi-channel constant-current drivers for the different lighting functions.

A central microcontroller orchestrates the system, managing CAN commands and enabling capabilities such as seamless switching between lighting modes, real-time diagnostics, and temperature-based derating or protection. It can detect fault conditions, such as open or short circuits, and communicate them over the CAN or LIN network. This transforms lighting from a passive function into an intelligent, networked subsystem.

From a system perspective, this creates a highly scalable and efficient platform. Instead of designing separate LED driver hardware for each model, manufacturers can adopt a unified, programmable architecture. Current levels, voltage